

AI Response Quality Evaluator

Batch Evaluation Report

This report presents an automated quality assessment of AI-generated responses using a multi-agent evaluation framework. Each response is independently analyzed across four quality dimensions before producing an overall verdict.

Generated On	13 August 2026, 10:00 AM
Report Type	Batch Evaluation
Evaluation Engine	Multi-Agent LLM Evaluator
Framework	Retrieval-Augmented Generation (RAG)
LLM Provider	Groq

Total Responses	Average Score
15	7.7/10
Pass	Needs Improvement
6	6

Evaluation Dimensions

- ✓ Relevance
- ✓ Accuracy
- ✓ Hallucination Detection
- ✓ Completeness

This report was generated automatically by the AI Response Quality Evaluator.

Executive Summary

This report presents the overall performance of the AI Response Validation System with Hallucination Detection Assistance. The submitted responses were evaluated using multiple specialized agents to measure relevance, accuracy, hallucination detection and completeness before generating an overall quality verdict.

Batch Statistics

Metric	Value
Total Responses	15
Successfully Evaluated	15
Average Score	7.70/10
Pass	6
Needs Improvement	6
Fail	3

Dimension-wise Performance

Dimension	Average Score
Relevance	7.6/10
Accuracy	7.3/10
Hallucination	8.2/10
Completeness	7.7/10

Key Findings

- Average Overall Score: **7.70/10**
- Best Performing Dimension: **Hallucination**
- Weakest Dimension: **Accuracy**
- Hallucination Frequency: **60.0%**
- Highest Response Score: **9.8/10**
- Lowest Response Score: **2.5/10**

Overall Assessment

The evaluated responses demonstrate satisfactory overall quality. While most responses are relevant and factually reasonable, improvements are recommended in weaker evaluation dimensions to further increase response reliability.

Batch Analytics

Metric	Value
Average Relevance	7.6/10
Average Accuracy	7.3/10
Average Hallucination	8.2/10
Average Completeness	7.7/10
Highest Overall Score	9.8/10
Lowest Overall Score	2.5/10
Best Performing Dimension	Hallucination
Weakest Dimension	Accuracy
Hallucination Frequency	60.0%
Grounded Responses	6

Verdict Distribution

Verdict	Count	Percentage
Pass	6	40.0%
Needs Improvement	6	40.0%
Fail	3	20.0%

Summary

A total of **15** responses were processed. **15** responses completed successfully. The average overall evaluation score was **7.7/10**. The strongest evaluation dimension was **Hallucination**, while the weakest was **Accuracy**. The detected hallucination frequency across the successfully evaluated responses was **60.0%**.

Individual Evaluation Results

Response #1

Question

What is Machine Learning?

AI Response

Machine Learning is a branch of Artificial Intelligence that enables computers to learn patterns from data and improve performance without being explicitly programmed.

Reference Answer

Machine Learning is a subset of Artificial Intelligence that allows systems to learn from data and make predictions or decisions without explicit programming.

Dimension	Score
Relevance	9/10
Accuracy	8/10
Hallucination	10/10
Completeness	9/10
Overall	8.9/10

Pass

Response #2

Question

What is the capital of France?

AI Response

The capital of France is Paris.

Reference Answer

Paris is the capital city of France.

Dimension	Score
Relevance	9/10
Accuracy	10/10
Hallucination	10/10
Completeness	10/10
Overall	9.8/10

Pass

Response #3

Question

What is photosynthesis?

AI Response

Photosynthesis is the process by which green plants use sunlight, carbon dioxide, and water to produce glucose and oxygen.

Reference Answer

Photosynthesis is the biological process in which plants convert sunlight, carbon dioxide, and water into glucose and oxygen.

Dimension	Score
Relevance	9/10
Accuracy	9/10
Hallucination	7/10
Completeness	9/10
Overall	8.5/10

Needs Improvement

Unsupported Claims

- Photosynthesis is the process by which green plants use sunlight, carbon dioxide, and water to produce glucose and oxygen.

Response #4

Question

What is the speed of light?

AI Response

The speed of light is approximately 299,792 kilometers per second in a vacuum.

Reference Answer

The speed of light in vacuum is approximately 299,792 km/s.

Dimension	Score
Relevance	9/10
Accuracy	10/10
Hallucination	10/10
Completeness	9/10
Overall	9.6/10

Pass

Response #5

Question

Who invented the telephone?

AI Response

Alexander Graham Bell is widely credited with inventing the first practical telephone.

Reference Answer

Alexander Graham Bell patented the first practical telephone in 1876.

Dimension	Score
Relevance	9/10
Accuracy	9/10
Hallucination	7/10
Completeness	9/10
Overall	8.5/10

Needs Improvement

Unsupported Claims

- Alexander Graham Bell is widely credited with inventing the first practical telephone.

Response #6

Question

What is Python?

AI Response

Python is a popular high-level programming language used for web development, artificial intelligence, data science, automation, and software development.

Reference Answer

Python is a versatile high-level programming language known for readability and extensive libraries.

Dimension	Score
Relevance	9/10
Accuracy	8/10
Hallucination	7/10
Completeness	9/10
Overall	8.2/10

Needs Improvement

Unsupported Claims

- Python is a popular high-level programming language used for web development, artificial intelligence, data science, automation, and software development.

Response #7

Question

What is cloud computing?

AI Response

Cloud computing provides computing resources such as servers, storage, databases, and software over the internet.

Reference Answer

Cloud computing delivers computing services including servers, storage, databases, networking, and software over the internet.

Dimension	Score
Relevance	9/10
Accuracy	8/10
Hallucination	10/10
Completeness	9/10
Overall	8.9/10

Pass

Response #8

Question

What is blockchain?

AI Response

Blockchain is a decentralized digital ledger that records transactions securely across multiple computers.

Reference Answer

Blockchain is a distributed ledger technology that securely records transactions across a network.

Dimension	Score
Relevance	9/10
Accuracy	8/10
Hallucination	7/10
Completeness	9/10
Overall	8.2/10

Needs Improvement

Unsupported Claims

- Blockchain is a decentralized digital ledger that records transactions securely across multiple computers.

Response #9

Question

What is the largest planet?

AI Response

Jupiter is the largest planet in our Solar System.

Reference Answer

Jupiter is the largest planet in the Solar System.

Dimension	Score
Relevance	9/10
Accuracy	10/10
Hallucination	10/10
Completeness	9/10
Overall	9.6/10

Pass

Response #10

Question

What is data mining?

AI Response

Data mining is the process of discovering useful patterns, trends, and knowledge from large datasets using statistical and machine learning techniques.

Reference Answer

Data mining extracts meaningful patterns and knowledge from large datasets.

Dimension	Score
Relevance	9/10
Accuracy	8/10
Hallucination	7/10
Completeness	9/10
Overall	8.2/10

Needs Improvement

Unsupported Claims

- Data mining is the process of discovering useful patterns, trends, and knowledge from large datasets using statistical and machine learning techniques.

Response #11

Question

What is Artificial Intelligence?

AI Response

Artificial Intelligence is the simulation of human intelligence by machines that can perform tasks like reasoning, learning, perception, and decision making.

Reference Answer

Artificial Intelligence is the capability of machines to perform tasks requiring human intelligence such as learning, reasoning, and problem solving.

Dimension	Score
Relevance	9/10
Accuracy	8/10
Hallucination	7/10
Completeness	9/10
Overall	8.2/10

Needs Improvement

Unsupported Claims

- Artificial Intelligence is the simulation of human intelligence by machines that can perform tasks like reasoning, learning, perception, and decision making.

Response #12

Question

Who discovered gravity?

AI Response

Gravity was discovered by Albert Einstein after he developed the theory of relativity.

Reference Answer

Isaac Newton formulated the law of universal gravitation after observing a falling apple.

Dimension	Score
Relevance	1/10
Accuracy	2/10
Hallucination	7/10
Completeness	1/10
Overall	2.8/10

Fail

Unsupported Claims

- Gravity was discovered by Albert Einstein after he developed the theory of relativity.

Response #13

Question

What is the capital of Australia?

AI Response

The capital of Australia is Sydney.

Reference Answer

Canberra is the capital city of Australia.

Dimension	Score
Relevance	1/10
Accuracy	1/10
Hallucination	7/10
Completeness	1/10
Overall	2.5/10

Fail

Unsupported Claims

- The capital of Australia is Sydney.

Response #14

Question

What is the boiling point of water?

AI Response

Water usually boils at around 50°C under normal atmospheric pressure.

Reference Answer

Water boils at approximately 100°C at standard atmospheric pressure.

Dimension	Score
Relevance	4/10
Accuracy	2/10
Hallucination	7/10
Completeness	4/10
Overall	4.2/10

Fail

Unsupported Claims

- Water usually boils at around 50°C under normal atmospheric pressure.

Response #15

Question

What is DNA?

AI Response

DNA stores genetic information used for growth and reproduction in living organisms.

Reference Answer

DNA (Deoxyribonucleic Acid) carries hereditary genetic information in almost all living organisms.

Dimension	Score
Relevance	9/10
Accuracy	8/10
Hallucination	10/10
Completeness	9/10
Overall	8.9/10

Pass

Flagged Responses

Response #3

Verdict: Needs Improvement

Overall Score: 8.5/10

Unsupported Claims

- Photosynthesis is the process by which green plants use sunlight, carbon dioxide, and water to produce glucose and oxygen.

Response #5

Verdict: Needs Improvement

Overall Score: 8.5/10

Unsupported Claims

- Alexander Graham Bell is widely credited with inventing the first practical telephone.

Response #6

Verdict: Needs Improvement

Overall Score: 8.2/10

Unsupported Claims

- Python is a popular high-level programming language used for web development, artificial intelligence, data science, automation, and software development.

Response #8

Verdict: Needs Improvement

Overall Score: 8.2/10

Unsupported Claims

- Blockchain is a decentralized digital ledger that records transactions securely across multiple computers.

Response #10

Verdict: Needs Improvement

Overall Score: 8.2/10

Unsupported Claims

- Data mining is the process of discovering useful patterns, trends, and knowledge from large datasets using statistical and machine learning techniques.

Response #11

Verdict: Needs Improvement

Overall Score: 8.2/10

Unsupported Claims

- Artificial Intelligence is the simulation of human intelligence by machines that can perform tasks like reasoning, learning, perception, and decision making.

Response #12

Verdict: Fail

Overall Score: 2.8/10

Unsupported Claims

- Gravity was discovered by Albert Einstein after he developed the theory of relativity.

Response #13

Verdict: Fail

Overall Score: 2.5/10

Unsupported Claims

- The capital of Australia is Sydney.

Response #14

Verdict: Fail

Overall Score: 4.2/10

Unsupported Claims

- Water usually boils at around 50°C under normal atmospheric pressure.

Improvement Recommendations

- Improve factual correctness and verify important claims before generating responses.
- Hallucination frequency is relatively high. Increase retrieval grounding and reference verification.

End of Report

Generated by
AI Response Quality Evaluator